



Attaching the PPT that got us selected for the SIH 2025

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SMART INDIA HACKATHON 2025



Problem Statement ID - 25184

Problem Statement Title - Secure closed group communication platform over existing public mobile communication networks for defence personnel and families.

Theme - Blockchain & Cybersecurity

PS Category - Software

Team ID - 72808

Team Name - INNOVISION



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RAKSHA-SETU



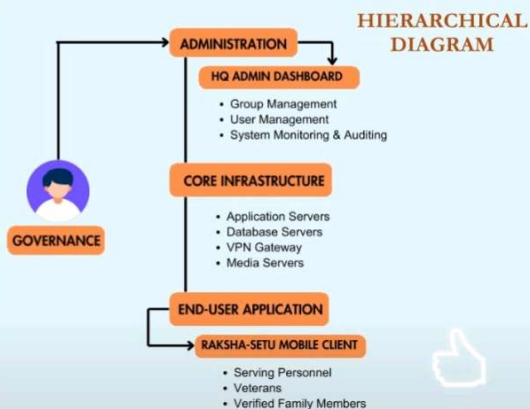
A SECURE COMMUNICATION ECOSYSTEM FOR THE MODERN MILITARY COMMUNITY

PROBLEM OVERVIEW

- Defence personnel using commercial apps risk data leaks and espionage.
- Foreign-hosted platforms expose sensitive defence communication.
- No sovereign, secure app exists for defence-exclusive communication.
- Commercial apps lack strict access control and data leak prevention.
- Sensitive family and operational details remain vulnerable on public platforms.

PROBLEM SOLUTION

- **HQ-Verified Closed Groups** – Only verified serving personnel, veterans, and registered families can access.
- **Mandatory VPN Encapsulation** – All communication routed through a defence-controlled VPN gateway.
- **End-to-End Encryption** – Messages, calls, and media remain fully encrypted.
- **Data Leak Prevention (DLP)** – Prevents screenshots, screen recording, copy-paste, and message forwarding.
- **On-Premise Hosting in India** – Ensures data sovereignty under military oversight.
- **Multi-modal Communication** – Secure text, multimedia sharing, and voice/video calling.



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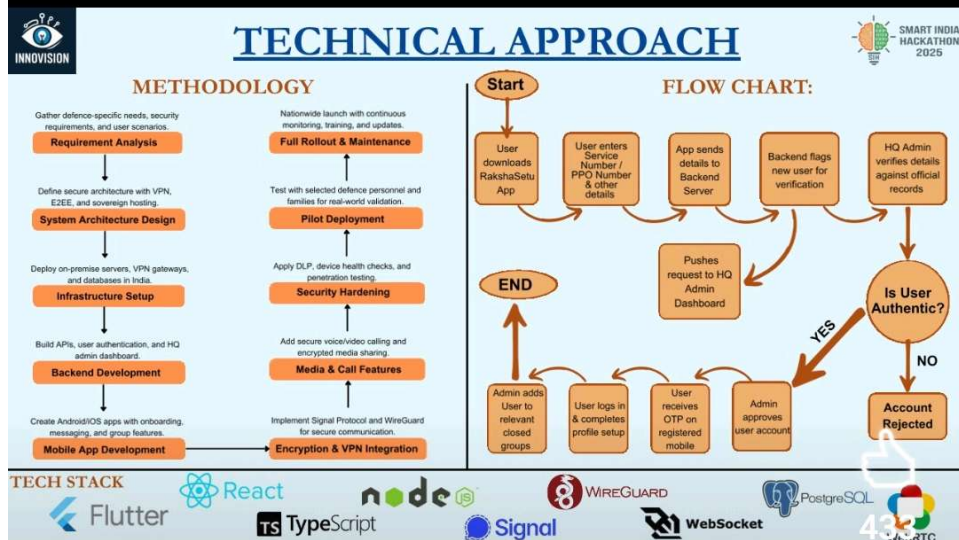


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FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY

Robust encryption (Signal Protocol), VPN encapsulation (WireGuard), and secure on-premise infrastructure are achievable using mature technologies and modern backend frameworks.

2. ECONOMICAL FEASIBILITY

While initial setup requires high investment in secure infrastructure, long-term costs are manageable through controlled scaling, microservices, and military-operated servers, ensuring high value for national security.

3. SOCIAL FEASIBILITY

The platform builds trust among defence personnel, veterans, and families by ensuring safe communication, strengthening morale and reducing risks of information leakage.

4. LEGAL & COMPLIANCE FEASIBILITY

All data is hosted within Indian territory, ensuring compliance with data sovereignty, CERT-In, and government cybersecurity frameworks, making it fully legitimate and acceptable.

5. OPERATIONAL FEASIBILITY

A user-friendly mobile app with guided onboarding ensures easy adoption, while centralized HQ admin controls simplify group management, verification, and monitoring.

6. SECURITY FEASIBILITY

Continuous penetration testing, DevSecOps pipeline, and strict DLP measures ensure resilience against cyberattacks, insider threats, and device-level compromises.

POTENTIAL CHALLENGES AND RISKS

SECURITY VULNERABILITIES

Risk of cyberattacks from state-sponsored actors targeting servers, VPN, or the app.



USER ADOPTION & TRAINING

Defence personnel and families may resist shifting from familiar apps to a new platform.



PERFORMANCE & SCALABILITY

Ensuring smooth operation under high load, especially in low-bandwidth or remote areas.



INSIDER THREATS

Malicious actions from authorized users with legitimate access.



STRATEGIES FOR OVERCOMING THESE CHALLENGES

SECURITY HARDENING

Regular penetration testing, DevSecOps pipeline, MFA integration, and third-party security audits.



AWARENESS & TRAINING

Simple UI/UX design, training modules, and campaigns to highlight risks of commercial apps.



OPTIMIZED ARCHITECTURE

Microservices design, load balancing, efficient VPN protocols like WireGuard, and offline-first features.



ACCESS CONTROL & MONITORING

Enforce least privilege principle, maintain immutable audit logs, and enable real-time anomaly detection.



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IMPACT AND BENEFITS



UNIQUE FEATURES:

Mandatory VPN Encapsulation

All traffic passes through a defence-controlled VPN.

HQ-Centric Trust Model

Users and groups are verified and managed only by Defence HQ.

Total Information Containment

No data leaves the secure ecosystem.

Defence Community Exclusivity

Accessible only to personnel, veterans, and families.

Data Sovereignty Guarantee

All infrastructure hosted within India under Government Authority.

IMPACTS

Enhanced National Security



Closes a major vulnerability by preventing hostile intelligence from exploiting communication channels.

Improved OPSEC



Reduces accidental leakage of troop movements, locations, and routines.

Family Protection



Shields families from being targeted for sensitive information or coercion.

Digital Sovereignty



Eliminates reliance on foreign platforms, strengthening India's cyber independence.

Resilient Defence Communication



Builds a secure backbone for both personal and operational communication.

BENEFITS

Trustworthy Ecosystem



Provides a safe digital environment exclusively for the defence community.

High Morale



Boosts confidence among personnel knowing their communication is secure.

Efficient Monitoring



HQ-admin tools simplify verification, group control, and auditing.

Scalable Solution



Architecture supports growth from small groups to thousands of users.

Long-Term Cost Value



Though, high set up cost, it ensures lasting security worth the investment.



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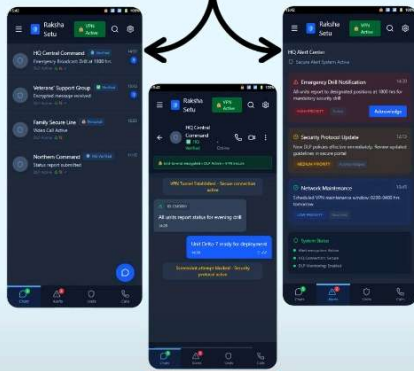
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OUR PROTOTYPE

App Interface



Chat Interface

RESEARCH AND REFERENCES



• Cryptographic Protocols for Secure Messaging

Summary: Explains the Signal Protocol, used in apps like Signal and WhatsApp, providing end-to-end encryption, forward secrecy, and resistance to surveillance.

Link: <https://signal.org/docs/>

• VPN Security & Performance Evaluation

Summary: Introduces WireGuard, a modern VPN protocol with superior speed, security, and simplicity compared to IPsec and OpenVPN, making it ideal for sensitive communication.

Link: <https://www.wireguard.com/papers/wireguard.pdf>

• Mobile Security & Data Leak Prevention

Summary: Defines major mobile app vulnerabilities and provides best practices for secure coding, DLP (screenshot blocking, copy-paste prevention), and device-level protection.

Link: <https://owasp.org/www-project-mobile-top-10/>

• Risks of Using Commercial Apps for Defense Communication

Summary: Using commercial apps for defense communication risks data leak and unauthorized access due to weak encryption and controls.

Link: <https://owasp.org/www-project-mobile-top-10/>

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Low Cost Machinery for Picking of Cotton

- Problem Statement ID – 25266
- Problem Statement Title -
Low Cost Machinery for Picking of Cotton
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Hardware
- Team Name (Registered on portal) Mindsmiths



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PROPOSED SOLUTION



Problem at Hand

Labor Intensive: Manual cotton picking accounts for 30-40% of production costs and requires long hours of repetitive work.

Rising Labor Shortage: Rural labor availability is declining, making cotton harvesting expensive and unreliable.

Mechanical Limitations: Industrial cotton harvesters cost ₹4-5 lakh, are heavy, and cannot operate in small or uneven fields typical in India.

Quality Challenges: Manual picking often results in higher trash content (>15%), reducing lint quality and market value.

Our Solution [EcoPick Web Demo](#)



Why We Stand Out

- **Affordable & Accessible:** Complete smart cotton picker under ₹10,000, ideal for small and marginal farmers.
- **AI + IoT Precision:** Picks only mature bolls, monitors crop & soil health, and provides real-time analytics.
- **Portable & Safe:** Lightweight, ergonomic, solar-hybrid powered, with protective features for operator safety.

Key Features

- **AI Picking:** Detects and harvests only mature cotton bolls.
- **Real-time soil, crop, and yield monitoring:** IoT Analytics
- **Lightweight backpack suitable for all operators:** Ergonomic Design
- **Vibration & current sensors for early fault alerts:** Predictive Maintenance
- **Anti-static sack with multi-stage dust filter:** Clean Harvesting
- **GPS mapping, yield and productivity tracking:** GPS mapping, yield and productivity tracking



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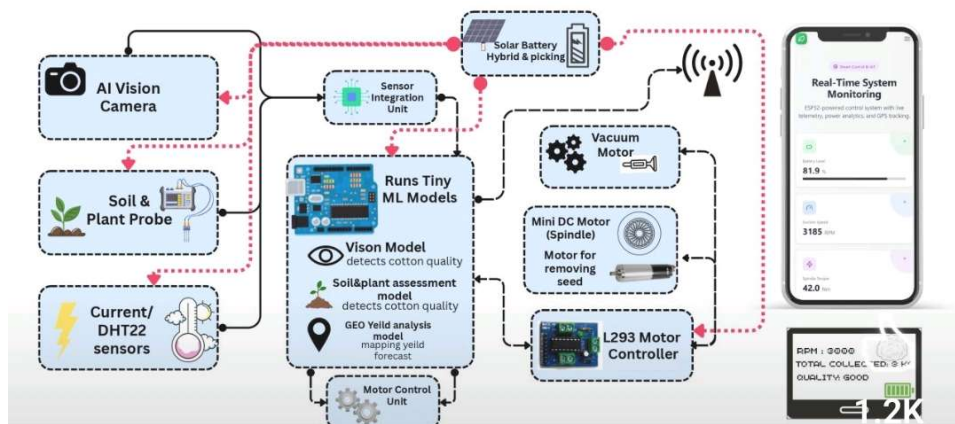
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TECHNICAL APPROACH



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FEASIBILITY AND VIABILITY



Technical Feasibility

- Built using commercially available components (ESP32-C3, Li-ion battery, solar panel, vacuum motor, 3D-printed frame).
- AI insights & IoT-enabled Wi-Fi connectivity for real-time soil, pH, and yield data sync.
- Low-power sensors (NPK, moisture, pH) integrated without stressing battery life.

Economic and Operational Viability

- Total device cost under ₹10,000, with daily operational cost < ₹40.
- Payback within 1-2 harvests for 1-2 acre farms; ROI > 40%.
- Lightweight backpack design reduces operator fatigue by 30-40%.
- Hip-mounted controller & multilingual mobile app.

Economical Feasibility

Evaluation Parameter	Current Manual / Portable Picker	Proposed Smart System
Cost of Device	₹4,000 - ₹7,000	₹9,500 - ₹10,000
Manpower Required	2-3 workers	1 operator
Daily Picking Capacity	~17-25 kg	30-35 kg (selective, clean lint)
Trash Content	10-15%	6-8%
Labour Cost Reduction	—	~80%
Power Consumption	Battery only (limited)	Solar-battery hybrid (sustainable)

Implementation Viability

Factor	Support Element
Manufacturability	Simple low-cost machining or rural fab-lab assembly possible.
Component Availability	100% components available in India.
Serviceability	Easy component replacement; no dependency on sophisticated tools.
Scalability	Modular add-ons (multi-nozzle picker, fiber sensor integration).
Deployment Readiness	Prototype ready for field testing; estimated commercial scaling possible within 12 months.

Maintenance & Safety Feasibility

- Minimal maintenance – few moving parts, easy to service with basic hand tools.
- Electrostatic shielding, dust filters, and insulated motor housing ensure user safety.
- Predictive maintenance via vibration and current sensing extends machine life.

Environmental Viability

- Low-heat, anti-static materials meet agricultural safety standards.
- Promotes climate-resilient and low-carbon farming practices.
- Supports Atmanirbhar Bharat & Digital India missions through digital literacy and rural innovation.



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RESEARCH AND REFERENCES



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- 2) J. Bankar, G. Nimbalkar, M. Nimbalkar, A. Hon, S. Ghodke and S. Tamboli, "Design & Development of Smart Cotton-Picking Robot Using ML & Suction Mechanism," 2025 SAE Media Group TechBriefs Contest - Robotics & Automation, June 1 2025.
- 3) C. Zhang et al., "The Development of Wear Characteristics of the Picking Hook Teeth," Wear, vol. XXXX, 2024. doi:10.1016/j.wear.2024.00060.

- 4) A. D. Abdazimov, S. S. Radjabov and N. N. Omonov, "Automation of agrotechnical assessment of cotton harvesting machines," Journal of Physics: Conference Series, vol. 1260, no. 3, p. 032001, 2019. doi:10.1088/1742-6596/1260/3/032001.

LINKS

- | | |
|------------------------------|-------------------------------|
| 1) Documentation | 4) Budget Report |
| 2) TAM SAM SOM Documentation | 5) Demo Video Link - Web Demo |
| 3) Blender Model Images | 6) Actual Website |



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